Ch. 5 CW Pt. 2 – Solns

Principles of Microeconomics

September 21, 2025

1. (a) Increase – the introduction of additional substitutes will increase the substitutability of Ford SUVs. That should increase the price elasticity of demand.

   (b) Decrease – the removal of substitutes will decrease the substitutability of Ford SUVs. That should decrease the price elasticity of demand.

   (c) Increase – if people think that SUVs are uniquely safe, that will decrease their substitutability, so the price elasticity of demand should increase.

   (d) Increase – a longer time horizon gives people more time to find substitutes and companies more time to introduce substitutes, such as the four-wheel-drive cargo van. That should increase the price elasticity of demand.

2. (a)

$$\eta = \left| \frac{\%\Delta Q_D}{\%\Delta P} \right|$$

$$\%\Delta Q = \frac{6-5}{\frac{6+5}{2}} = \frac{2}{11}$$

$$\%\Delta P = \frac{4-5}{\frac{4+5}{2}} = \frac{-2}{9}$$

$$\eta = \left| \frac{\frac{2}{11}}{\frac{-2}{9}} \right| = \frac{9}{11}$$

   (b) $\eta < 1$, so demand for gas is inelastic.

   (c) Demand would be more elastic. A longer time horizon gives people more time to find substitutes or change their behavior so that they consume a different amount of gas. As a result, gas becomes more substitutable, and the price elasticity of demand increases.

3. (a)

$$PES = \frac{\%\Delta Q_S}{\%\Delta P}$$
$$1.6 = \frac{\%\Delta Q_S}{-0.18}$$
$$-0.288 = \%\Delta Q_S$$

Q_S decreases by 28.8%.

(b)

$$IED = \frac{\%\Delta Q_D}{\%\Delta I}$$
$$0.6 = \frac{\%\Delta Q_D}{\frac{42{,}000-38{,}000}{40{,}000}}$$
$$0.6 = \frac{\%\Delta Q_D}{0.1}$$
$$0.06 = \%\Delta Q_D$$

Q_D increases by 6%.

(c) The cross-price elasticity of demand would be negative because the two goods are complements. An increase in the price of iPhones will decrease the quantity demanded of portable chargers.

(d) Let $x = \%\Delta Q_D$.

$$TR = PQ$$
$$1.02TR = 1.05P * (1+x)Q$$
$$1.02TR = 1.05(1+x)PQ$$
$$1.02 = 1.05(1+x)$$
$$\frac{1.02}{1.05} - 1 = x$$
$$-0.03 = x$$
$$\eta = \left|\frac{-3\%}{5\%}\right| = 0.6$$